

DOCUMENT #1 OF 10 — PROJECT BLUEPRINT

Product Requirement Document (PRD)

AI-Powered Intelligent Resume Screening & Candidate Assessment Platform

Product Name	Document Version	Author / Owner	Status
ScreenAI Pro	v1.0 (Production Target)	Lead AI Architect & Product Lead	Approved for Implementation

1. Executive Summary & Problem Statement

Modern talent acquisition teams face overwhelming volumes of applications for open engineering and corporate roles, with top listings regularly receiving hundreds or thousands of resumes within 48 hours. Human recruiters spend an average of 6 to 8 seconds manually scanning each resume, leading to high fatigue, unconscious bias, missed high-potential candidates, and prolonged hiring cycles exceeding 45 days.

CORE PROBLEM STATEMENT

Traditional Keyword-based Applicant Tracking Systems (ATS) reject qualified candidates due to minor formatting discrepancies or semantic differences (e.g., missing exact synonym keywords), while human screening cannot scale efficiently. There is a lack of an automated end-to-end platform that combines deep contextual parsing, multi-candidate ranking, ATS compliance scoring, automated technical skill verification, and interactive virtual interview assessments into a single unified recruiter workspace.

2. Product Vision & Goals

- **Intelligent Contextual Matching:** Move beyond rigid keyword matching to semantic NLP understanding of competencies, project complexities, and career progression.
- **Recruiter Efficiency Multiplier:** Reduce initial resume screening time by over 80% while generating automated candidate ranking and gap analyses.
- **End-to-End Evaluation Funnel:** Unify screening, duplicate detection, ATS compliance audit, technical round evaluations, and AI-conducted virtual interviews in one platform.
- **Actionable Candidate Feedback:** Provide constructive resume optimization insights for candidates and deep-dive technical gap reports for recruiters.

3. Target Audience & Personas

Persona A: Corporate Recruiters & HR Talent Teams

Needs to process 500+ resumes per job post, auto-rank candidates against strict and optional requirements, filter by experience tiers, and export shortlists with AI justification notes.

Persona B: Engineering Hiring Managers & Interviewers

Requires objective technical verification, candidate project depth breakdown, automated technical round assessments, and transcripts from AI-guided virtual interviews.

4. Comprehensive Core Feature Breakdown

Module	Feature Description & Scope	Input / Output	Priority
1. Virtual Interview Agent	Interactive AI interviewer conducting real-time technical and behavioral video/audio/text interviews based on parsed resume claims and job criteria.	Audio/Video/Text Input → Transcript, Evaluation Score & Sentiment	P1 - High
2. Technical Round Evaluation	Automated technical coding/problem-solving and conceptual quiz generation tailored to candidate's stated skill level.	Live Code/Quiz Answers → Automated Rubric Score & Code Quality	P1 - High
3. AI Resume Parsing Engine	Contextual NLP extraction of contact details, education, work timeline, technical skill taxonomy, certifications, and project metrics from PDF/DOCX.	PDF/DOCX Resumes → Structured JSON Schema Entity Tree	P0 - Critical
4. Job-Resume Semantic Matcher	Calculates exact match %, semantic proximity score, identifies mandatory vs. optional skill overlaps, and highlights critical skill gaps.	Parsed Resume + Job Description → Match Score (0-100%) & Matrix	P0 - Critical
5. Candidate Ranking Matrix	Dynamically orders all applicants for a given job posting using composite weighted scoring (skills 50%, experience 30%, education 20%).	Candidate Pool → Global Rank 1..N with percentile metrics	P0 - Critical
6. ATS Compatibility Audit	Analyzes resume formatting, section header standard compliance, typography, column layouts, table artifacts, and text-extractability.	Raw Resume File → ATS Score (0-100) & Warning Flags	P0 - Critical
7. Duplicate Detection	Detects re-applications, identical or plagiarized resume content across candidates using fuzzy text hashing and unique biometric/email hashes.	Uploaded Resumes → Similarity Index (%) & Duplicate Flag	P1 - High
8. Resume Improvement Guide	Generates candidate-facing actionable feedback: missing high-value keywords, weak action verbs, metric quantification tips, and format repairs.	Parsed Resume → Step-by-step optimization recommendations	P2 - Medium
9. Recruiter Workspace & Triage	Interactive pipeline management: Star/Shortlist, Reject, Review Later, custom notes, bulk actions, and email dispatch triggers.	Recruiter Action → Pipeline State Update & Candidate Logs	P0 - Critical
10. Multi-Attribute Filter & Search	Instant parametric search across candidate pool by skill tags, years of experience, university tier, location, and match score ranges.	Filter Criteria Query → Real-time filtered candidate table	P0 - Critical

5. Success Metrics & Key Performance Indicators (KPIs)

- **Time-to-Shortlist Reduction:** Reduce recruiter manual screening time from 20 hours/job to under 45 minutes (95% decrease).

- **Parsing Accuracy Rate:** Maintain >96% entity extraction accuracy across diverse PDF formatting layouts and multi-column templates.
- **Semantic Ranking Reliability:** >90% recruiter agreement rate with top-10 auto-ranked candidate recommendations.
- **System Throughput:** Bulk parsing rate of ≥ 100 resumes per minute with concurrent processing queues.
- **ATS False Rejection Rate:** Lower candidate parsing failure to under 1.5%.

6. User Journey & Core Functional Workflow

1. **Job Creation:** Recruiter enters job title, experience bracket, mandatory skills (e.g., Python, Docker), and nice-to-have skills.
2. **Bulk Upload:** Recruiter drags and drops 100+ PDF/DOCX resumes simultaneously.
3. **AI Processing Pipeline:** The system parses entities → runs duplicate check → evaluates ATS score → performs vector-based semantic matching.
4. **Dashboard Review:** Recruiter views the ranked leaderboard (Rank 1 to N), filters by score (>80%), and inspects detailed candidate breakdown cards (strengths vs. missing skills).
5. **Action & Assessment:** Recruiter shortlists top candidates and triggers automated Technical Round or Virtual AI Interview links directly.

7. Future Roadmap & Out-of-Scope (Phase 2+)

- **Phase 1 (Current Scope):** Core AI parsing, matching, ranking, ATS auditing, duplicate detection, candidate filtering, recruiter notes, and interview evaluation framework.
- **Phase 2 (Out of Scope for initial release):** Autonomous live video avatar interviewer with real-time lip sync, deep HRIS platform integrations (Workday, Greenhouse API sync), and multi-lingual voice synthesis.